

Lexical Simplification for Scientific Texts: Revisiting SARI in the Era of Modern LLMs

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Motivation

- Scientific text difficult for non-experts due to domain jargon
- SimpleText@CLEF '25 Task 1.1: Make biomedical scientific text more accessible

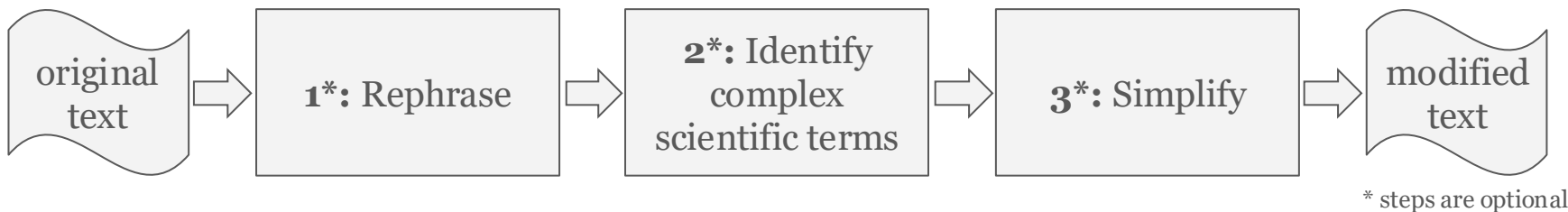
For every nine people treated with **haloperidol** instead of **olanzapine**, one fewer person would experience a...



For every nine people treated with **haloperidol** (a **medication for mental health conditions**) instead of **olanzapine** (another **mental health medication**), one fewer person would experience a...

Idea

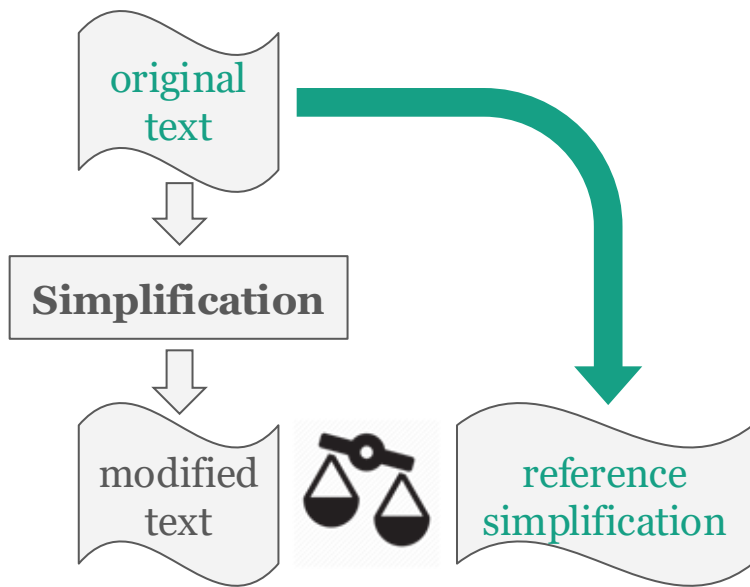
- Goal: Simplification for non-experts
→ knowledge of **language**, not of **biomedical** domain
- Low cost/small LLMs
- Find complex terms, simplify them, keep rest
- Prompts instructing LLMs to only explain/simplify complex terms



Submit + Results

- Submitted runs with different prompts, used cheap LLMs
- SimpleText '25 Task 1.1 mainly evaluated with SARI

SARI considers added, deleted, kept n-grams of produced against reference simplification



Model	P	f10	f1	orig	auto
baseline	-	-	-	7.844	12.033
gpt-4.1-nano	C	8	1	32.445	33.944
gemini-2.5	C	18	7	30.175	32.586
gemini-2.0	C	10	1	28.917	31.199
gemini-2.5	R	1	0	29.256	32.598
gemini-2.0	R	47	1	27.526	31.160
gpt-4.1-nano	P1	67	22	38.238	40.416
gemini-2.0	P1	42	22	25.858	30.001
gemini-2.0	C+P1 [△]	44	22	25.234	29.080
gpt-4.1-nano	P2	75	21	39.572	41.315
gemini-2.5	P2	4275	1550	25.062	27.811
gemini-2.0	P2	87	132	29.263	31.174
gemini-2.0	C+P2 [△]	91	132	27.561	30.024
gpt-4.1-nano	PN1	32	* < 320	35.262	37.262
gemini-2.0	PN1	174	1268	32.267	34.474
gemini-2.0	PI2	199	174	20.521	24.045

Runs
performed
okay
...but why?

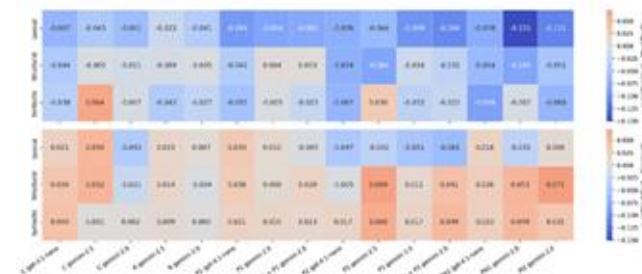
Is SARI still an adequate evaluation measure for LLM-based text simplification?

We considered different

- Numbers of complex terms
- LLMs
- Simplicity signals

Setting		Comprehensive				Targeted			
Model	P	f10	f1	original	auto	f10	f1	original	auto
gpt-4.1-nano	P1	72	17	37.484	39.896	67	22	38.238	40.416
gpt-4.1-nano	P2	60	24	39.135	40.647	75	21	39.572	41.315
gpt-4.1-mini	P1	95	20	26.623	30.271	129	33	26.288	30.121
gpt-4.1-mini	P2	109	30	29.388	32.223	126	39	29.336	31.475
gpt-5.4-nano	P1	51	19	25.803	29.608	62	15	26.806	29.852
gpt-5.4-nano	P2	38	13	27.138	30.323	44	14	27.731	31.025
gpt-5.4-mini	P1	66	20	22.338	26.362	63	20	21.957	25.948
gpt-5.4-mini	P2	65	16	33.107	35.312	79	15	34.010	36.480

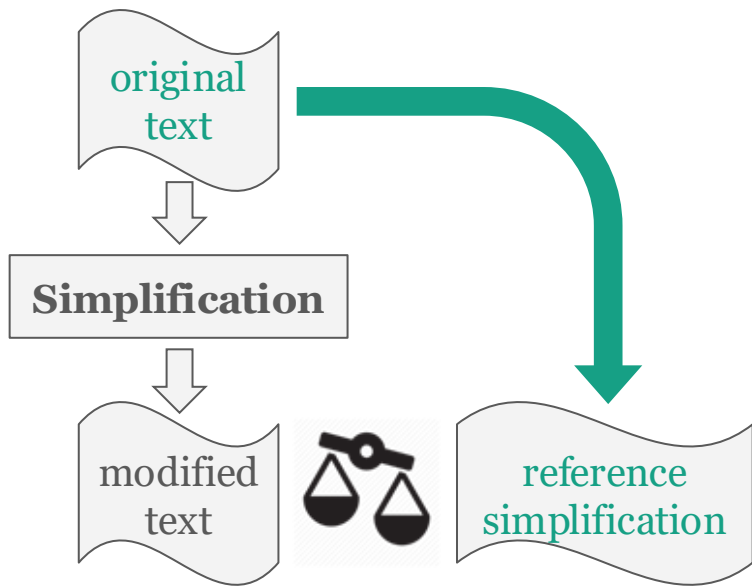
	θ	.01	.025	.05	.075	.1	.2	.3	.4	.5	1
# texts with CTI	7400	7285	6901	6500	5687	3824	2775	2566	2539	2524	



Model	P	f10	f1	orig	auto	FRE $\uparrow$	CL $\downarrow$	T $\downarrow$	B $\uparrow$
baseline	-	-	-	7.844	12.033	31.523	13.752	2.104	-
gpt-4.1-nano	C	8	1	32.445	33.944	-6.139	19.805	3.27	0.93
gemini-2.5	C	18	7	30.175	32.586	-14.528	20.887	3.015	0.927
gemini-2.0	C	10	1	28.917	31.199	-3.714	19.504	2.923	0.934
gemini-2.5	R	1	0	29.256	32.598	25.914	14.977	2.186	0.953
gemini-2.0	R	47	1	27.526	31.160	33.193	13.77 $\uparrow$	2.096$\uparrow$	0.843
gpt-4.1-nano	P1	67	22	38.238	40.416	34.498	13.308	2.726	0.928
gemini-2.0	P1	42	22	25.858	30.001	35.876	12.578	2.811	0.836
gemini-2.0	C+P1 Δ	44	22	25.234	29.080	36.001	12.585	2.815	0.948
gpt-4.1-nano	P2	75	21	39.572	41.315	31.238	13.829 $\uparrow$	2.816	0.923
gemini-2.5	P2	4275	1550	25.062	27.811	35.909	12.08	2.752	0.943
gemini-2.0	P2	87	132	29.263	31.174	35.759	12.674	3.003	0.829
gemini-2.0	C+P2 Δ	91	132	27.561	30.024	35.829	12.675	3.014	0.94
gpt-4.1-nano	PN1	32	* < 320	35.262	37.262	34.327	13.201	2.786	0.829
gemini-2.0	PN1	174	1268	32.267	34.474	43.769	11.475	2.674	0.842
gemini-2.0	PI2	199	174	20.521	24.045	36.925	11.966	2.479	0.953

Findings

- LLMs prompted to only modify complex components but changed syntax, structure & lexical features of text
- More powerful LLMs do not always achieve better SARI scores



More powerful LLMs
→ more rewriting
→ lower n-gram overlap
→ worse SARI

Worse SARI $\neq$ worse simplifications

Takeaways

- Be mindful what you measure
- SARI rewards conservative lexical edits but not rewriting

Big thanks to my student assistants
and the SimpleText organisers!

[illegible]